Mutual Fund Analytics Dashboard

CAPSTONE PROJECT REPORT

BLUESTOCK FINTECH INTERNSHIP

Prepared by **Lakshit Sharma** · Data Analytics Capstone · 2024

This report documents the end-to-end development of a Mutual Fund Analytics Dashboard — a comprehensive data analytics platform built during the Bluestock Fintech Internship. The project encompasses ETL pipeline development, SQLite database design, financial metric computation, interactive Streamlit dashboard creation, cloud deployment, and Markowitz Portfolio Optimization.

Certificate of Completion

Internship Completion Statement

This is to certify that **Lakshit Sharma** has successfully completed the Data Analytics Capstone Project as part of the **Bluestock Fintech Internship Program**. The project titled *Mutual Fund Analytics Dashboard* was designed, developed, and deployed in accordance with industry standards for data engineering and financial analytics.

The candidate demonstrated proficiency in Python programming, ETL pipeline architecture, SQL database design, financial metric computation, dashboard development using Streamlit, and cloud deployment — meeting all project objectives and deliverables.

Program

Bluestock Fintech Internship

Role

Data Analytics Intern

Project Type

Capstone — End-to-End

Status

✅ Successfully Completed

Acknowledgement

I would like to express my sincere gratitude to **Bluestock Fintech** for providing the opportunity to work on this challenging and rewarding capstone project. The internship offered invaluable practical exposure across the full spectrum of data analytics — from raw data extraction to production-grade dashboard deployment.

I am deeply thankful to my mentors and the internship program coordinators for their guidance on **ETL development**, **SQL database design**, **financial analysis methodologies**, and **cloud deployment best practices**. Their insights helped shape this project into a professional-grade analytics platform that meets real-world industry standards.

Executive Summary

Project Overview

The Mutual Fund Analytics Dashboard is an end-to-end data analytics platform developed during the Bluestock Fintech Internship. It addresses a critical gap in the Indian mutual fund ecosystem: the absence of a centralized, interactive tool that enables investors and analysts to compare funds, evaluate performance metrics, and make data-driven investment decisions.

Business Problem

Investors and financial analysts struggle to access consolidated mutual fund data. Information is scattered across multiple sources, manual calculations are error-prone, and there is no unified dashboard for performance comparison, risk analysis, or portfolio optimization.

Solution Developed

A fully automated analytics pipeline was built — from API data extraction through ETL processing, SQLite storage, financial metric computation, and an interactive Streamlit dashboard — deployed publicly on Streamlit Community Cloud with Markowitz Portfolio Optimization as a bonus feature.

Technologies

Python · Pandas · NumPy · SQLite · Plotly · Streamlit · Git · GitHub · MFAPI

Final Outcome

A production-ready analytics platform with live NAV data, 10+ financial metrics, interactive charts, and portfolio optimization — accessible via cloud deployment.

This project demonstrates full-stack data analytics competency: ETL engineering, database design, financial modeling, visualization, and cloud deployment.

Introduction & Problem Statement

What Are Mutual Funds?

A mutual fund is a professionally managed investment vehicle that pools capital from multiple investors to purchase a diversified portfolio of securities — including equities, bonds, and money market instruments. Each investor holds units proportional to their contribution, and the fund's value is measured by its **Net Asset Value (NAV)**.

NAV — Net Asset Value

The per-unit market value of a fund's assets minus liabilities. NAV is the primary benchmark for fund valuation and is computed daily by fund houses.

AUM — Assets Under Management

The total market value of all investments managed by a fund. AUM reflects fund size, investor confidence, and operational scale.

Risk Analysis

Measures such as volatility, Sharpe Ratio, and Maximum Drawdown help investors understand the risk-return profile of a fund relative to its benchmark.

Why Dashboards Help Investors

Interactive dashboards consolidate fragmented data into a single view, enabling rapid comparison, trend analysis, and informed decision-making without manual calculations.

Current Challenges

🔴 No Centralized Analytics

Mutual fund data is scattered across AMFI, fund house websites, and third-party portals — making consolidated analysis extremely difficult.

🔴 Manual Calculations

Financial metrics like Sharpe Ratio, Alpha, and Beta require complex formulas that are time-consuming and error-prone when done manually.

🔴 Time-Consuming Analysis

Comparing even a handful of funds across multiple dimensions can take hours without automated tooling and pre-computed metrics.

Project Objectives & Architecture

Project Objectives

01

Collect Mutual Fund Data

Extract NAV, performance, and AUM data programmatically using the MFAPI.

02

Build ETL Pipeline

Design automated extraction, transformation, and loading workflows with validation.

03

Store Data in SQLite

Create a normalized relational database with dimension and fact tables.

04

Compute Financial Metrics

Calculate 10+ metrics including Sharpe Ratio, Alpha, Beta, and Maximum Drawdown.

05

Build Streamlit Dashboard

Develop an interactive, professional UI with KPI cards, charts, and fund tables.

06

Deploy Application

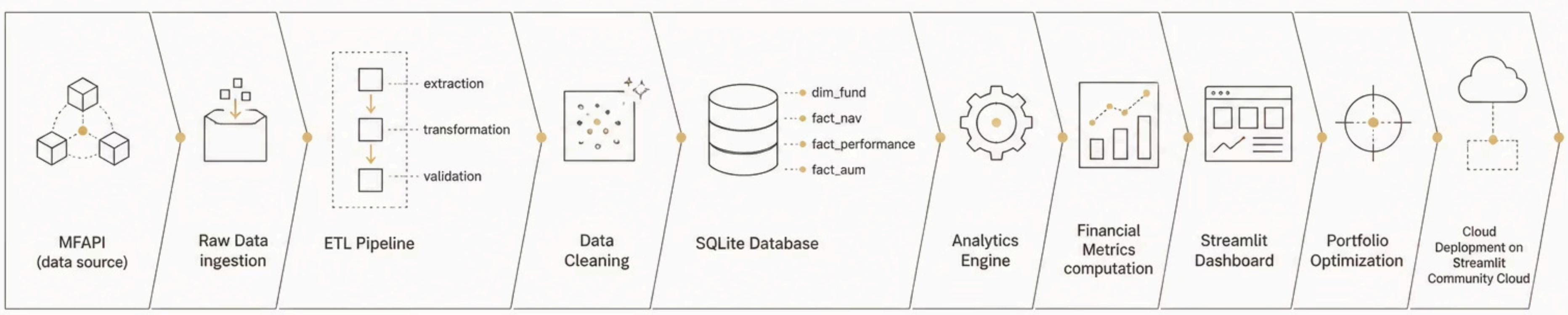
Host the dashboard publicly on Streamlit Community Cloud.

07

Portfolio Optimization

Implement Markowitz Modern Portfolio Theory with Efficient Frontier visualization.

Project Architecture



The architecture follows a modern data engineering pattern: raw data flows from the MFAPI through a validated ETL pipeline into a normalized SQLite database, where an analytics engine computes financial metrics that power the interactive Streamlit dashboard and portfolio optimization module — all deployed on the cloud for public access.

Technology Stack & Project Structure

Technology Stack

Dashboard

Category	Technology	Purpose
Language	Python 3.9+	Core programming language for all data processing and analytics
Data Manipulation	Pandas, NumPy	DataFrames, numerical computation, time-series analysis
Database	SQLite	Lightweight relational database for structured fund data storage
Visualization	Plotly	Interactive charts — line, bar, scatter, and area plots
Streamlit	Rapid web app framework for deploying analytics dashboards	
Version Control	Git, GitHub	Source control, collaboration, and public repository hosting
Data Source	MFAPI	Mutual Fund API for NAV, scheme info, and historical data
IDE	VS Code	Development environment with Python and SQL extensions

Project Folder Structure

```
dashboard/
├─── data/
│    ├─── raw/         # Raw API responses
│    ├─── processed/   # Cleaned CSVs
│    └─── db/          # SQLite database file
├─── scripts/
│    ├─── etl_pipeline.py
│    └─── metrics.py
├─── sql/
│    └─── schema.sql
├─── streamlit_app/
│    └─── app.py
├─── reports/
├─── README.md
└─── requirements.txt
```

Folder Purpose Reference

- **data/raw/** — Stores unprocessed JSON responses from MFAPI before any transformation
- **data/processed/** — Contains cleaned, normalized CSV files ready for database loading
- **data/db/** — Houses the SQLite database file (`mutual_funds.db`)
- **scripts/** — Python modules for ETL logic and financial metric computation
- **sql/** — Schema definition files for creating and migrating database tables
- **streamlit_app/** — Main dashboard application (`app.py`) and configuration
- **reports/** — Generated analytical reports and exported visualizations

Database Design & ETL Pipeline

Database Schema

The SQLite database follows a **star schema** design pattern with one dimension table and three fact tables, enabling efficient analytical queries across fund metadata, NAV history, performance data, and AUM trends.

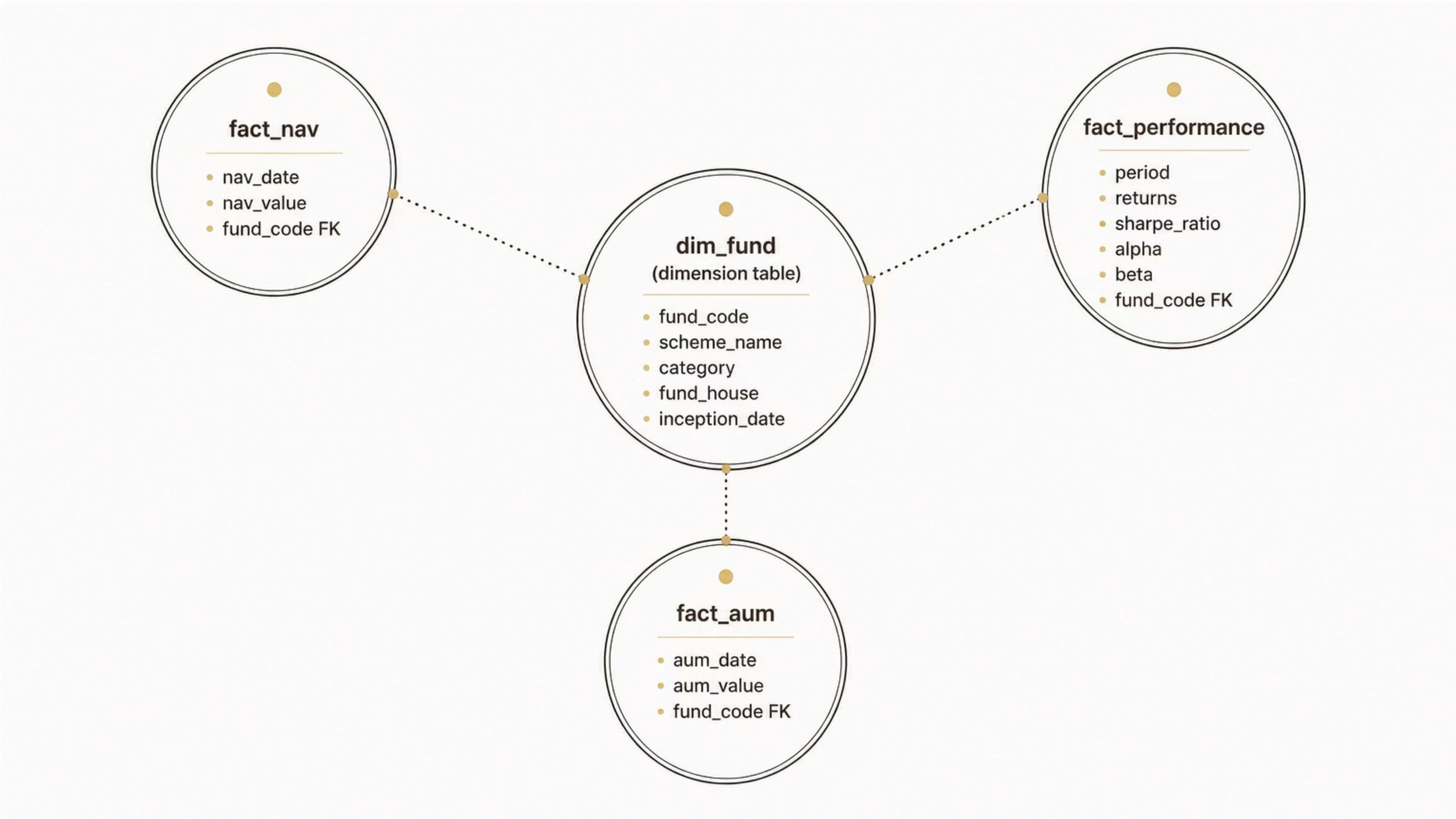
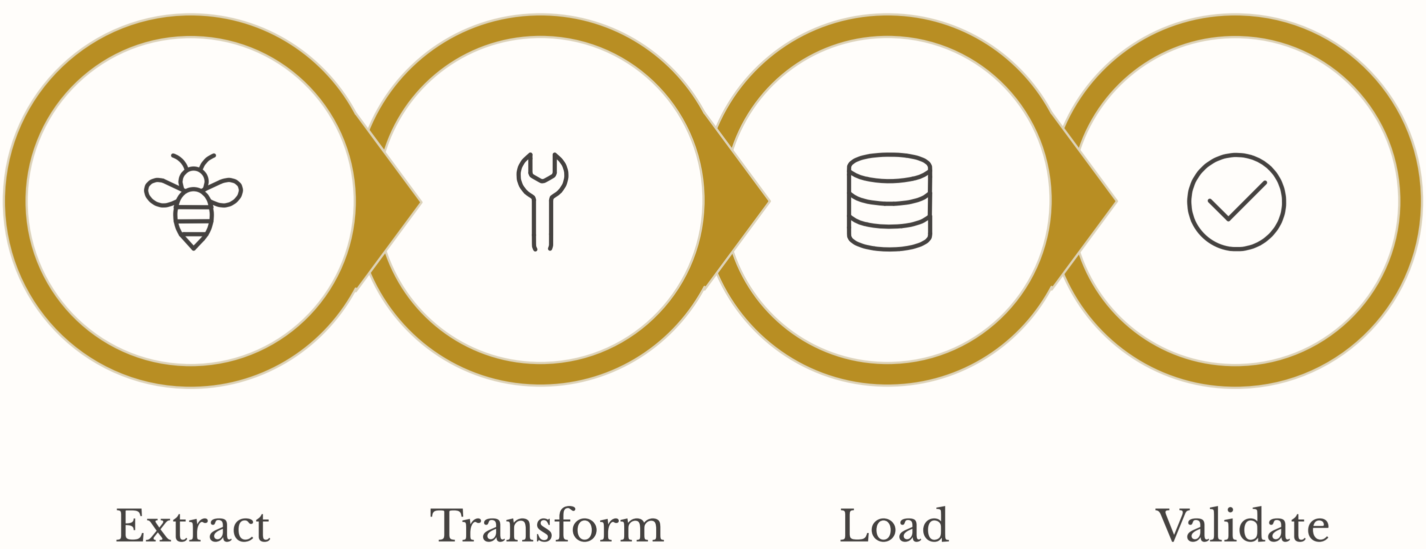


Table	Key Columns	Purpose
dim_fund	fund_code, scheme_name, category, fund_house	Master reference for all mutual fund schemes
fact_nav	nav_date, nav_value, fund_code (FK)	Daily NAV history for trend analysis
fact_performance	period, returns, sharpe_ratio, alpha, beta	Computed risk-adjusted performance metrics
fact_aum	aum_date, aum_value, fund_code (FK)	Assets Under Management over time

ETL Pipeline



The ETL pipeline automates the full data ingestion lifecycle. **Extraction** pulls scheme metadata and NAV history from MFAPI. **Transformation** handles missing values, date parsing, deduplication, and type casting. **Loading** writes clean records into SQLite using parameterized INSERT statements. **Validation** confirms row counts match, null thresholds are within limits, and foreign key relationships are intact before each run is marked successful.

Financial Metrics & Dashboard Features

Financial Metrics Computed

Return Metrics

- Annual Return — CAGR over selected period
- Alpha — Excess return vs. benchmark
- Beta — Sensitivity to market movements

Risk-Adjusted Metrics

- Sharpe Ratio — Return per unit of total risk
- Sortino Ratio — Return per unit of downside risk
- Volatility — Annualized standard deviation

Downside & Cost

- Maximum Drawdown — Largest peak-to-trough decline
- Expense Ratio — Annual management fee (%)
- Morningstar Rating — Star-based fund quality score

Scale Metrics

- Assets Under Management (AUM) — Total fund size
- NAV History — Daily unit price tracking
- Performance Table — Multi-period returns grid

Dashboard Features

KPI Cards

Top-level metrics displayed prominently — current NAV, AUM, 1-year return, Sharpe Ratio, and Morningstar rating for the selected fund.

Fund Selection & Master

Dynamic dropdown to select any mutual fund scheme. Fund Master table displays complete metadata including category, fund house, and inception date.

NAV History & Interactive Charts

Plotly-powered interactive charts showing NAV trends, return comparisons, and AUM growth — with zoom, pan, and hover tooltips.

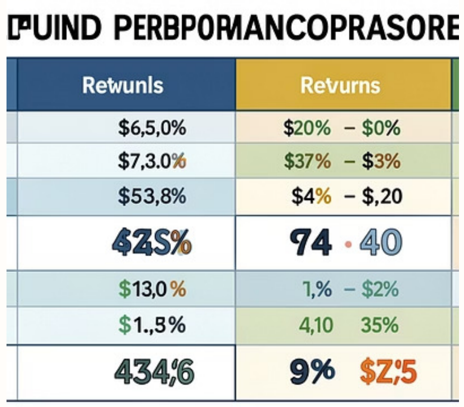
Performance Table

Structured table displaying returns across 1-month, 3-month, 6-month, 1-year, 3-year, and 5-year horizons for rapid fund comparison.

Dashboard Screenshots & Bonus Features

Dashboard Screenshots






Placeholders above represent: Dashboard Home, Fund Selection View, NAV Trend Chart, and Performance Metrics Table — all featuring a professional UI with consistent color palette and responsive layout.

Bonus Features

1

Professional Dashboard UI

A polished, corporate-grade interface with consistent typography, color-coded KPI cards, sidebar navigation, and responsive layout — designed to meet professional portfolio standards.

2

Cloud Deployment

Full deployment on **Streamlit Community Cloud**, making the dashboard publicly accessible. Deployment included configuring `requirements.txt`, setting up GitHub integration, and managing environment variables.

3

Interactive Analytics & KPI Visualization

Dynamic Plotly charts with drill-down capabilities, cross-filtering between components, and real-time metric recalculation based on user-selected fund and time period.

4

Markowitz Portfolio Optimization

Implements **Modern Portfolio Theory (MPT)** — constructing optimal portfolios by maximizing expected return for a given risk level. Features include **Efficient Frontier** visualization, **Sharpe Ratio optimization** to find the tangency portfolio, and weight allocation recommendations across selected funds.

Results, Challenges & Future Scope

Project Results

500+

Mutual Funds

Schemes indexed across equity, debt, hybrid, and liquid categories

100K+

NAV Records

Historical daily NAV data points stored and queryable

10+

Financial Metrics

Risk-adjusted and return metrics computed per fund

4

Database Tables

Normalized star schema with dimension and fact tables

1

Cloud Deployment

Live dashboard on Streamlit Community Cloud

1

Portfolio Optimizer

Markowitz MPT with Efficient Frontier visualization

Challenges Faced

→ **Deployment Issues**

Resolving dependency conflicts and environment configuration for Streamlit Cloud deployment.

→ **Data Cleaning**

Handling inconsistent date formats, missing NAV values, and duplicate scheme records from the API.

→ **ETL Pipeline Complexity**

Designing robust validation logic to ensure data integrity across thousands of fund records.

Future Scope

- **Machine Learning** — Predictive NAV modeling and fund recommendation engine
- **Live Market Data** — Real-time API integration for intraday updates
- **REST API Layer** — Expose analytics endpoints for third-party consumption
- **Power BI Integration** — Enterprise reporting and executive dashboards
- **User Authentication** — Personalized portfolios and saved watchlists
- **Email Reports** — Scheduled performance summaries delivered automatically
- **Monte Carlo Simulation** — Probabilistic return forecasting and scenario analysis